Technical Analysis using R

Development Phase

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Open this folder in RStudio and set it as the working directory. Run:

source("run_tests.R")

source("analyze_portfolio.R")

Or run each script with Rscript from this directory. No packages or internet are required.

The script reads the five Assignment 2 Yahoo Finance snapshots in market_data. The reference period is 2025; these are historical snapshots. The script writes one daily indicator CSV per symbol, a latest-date summary and a screen of stocks above SMA20 with RSI in [50, 70]. It uses daily Close and a 365-calendar-day filter relative to each file's last date. Buy is an upward EMA12/26 crossing with RSI below 70; Sell is a downward crossing. Signals are classroom examples.

Indicator conventions

SMA returns complete trailing windows. The analysis pads the leading dates with NA.

EMA starts at the first price. MACD returns the fast-minus-slow EMA, its signal EMA, and their difference.

Standard deviation divides by n (population).

Regression uses the handout's inclusive start/end indices, then fits local positions by least squares. At offset zero, regressionLength=20 spans 21 points where available; the analysis labels this parameter as 20.

RSI seeds Wilder averages from the first period price changes and emits the seed at period+1. Smoothing starts with the next change so the seed observation is counted once. A flat sequence scores 50.

Stochastic RSI follows the handout's full-series normalization and two SMA smoothers, with leading NA for alignment. A zero range maps to 0.5. The full-series range uses future observations relative to earlier rows, so neither stochastic line is used in the trading rules.

Crossover returns None/Up/Down. Crossunder returns None initially and True/False afterward, per the pseudocode. Missing comparisons do not fire signals.

ChatGPT (Codex) assisted with code adaptation, testing and documentation. Test and analysis logs record the assistant's R run.

Verified results

All nine indicators passed the reproducible R checks. Analysis produced 250 daily rows for each of five symbols. GOOGL and AMZN met the closing-date screen on 31 December 2025.